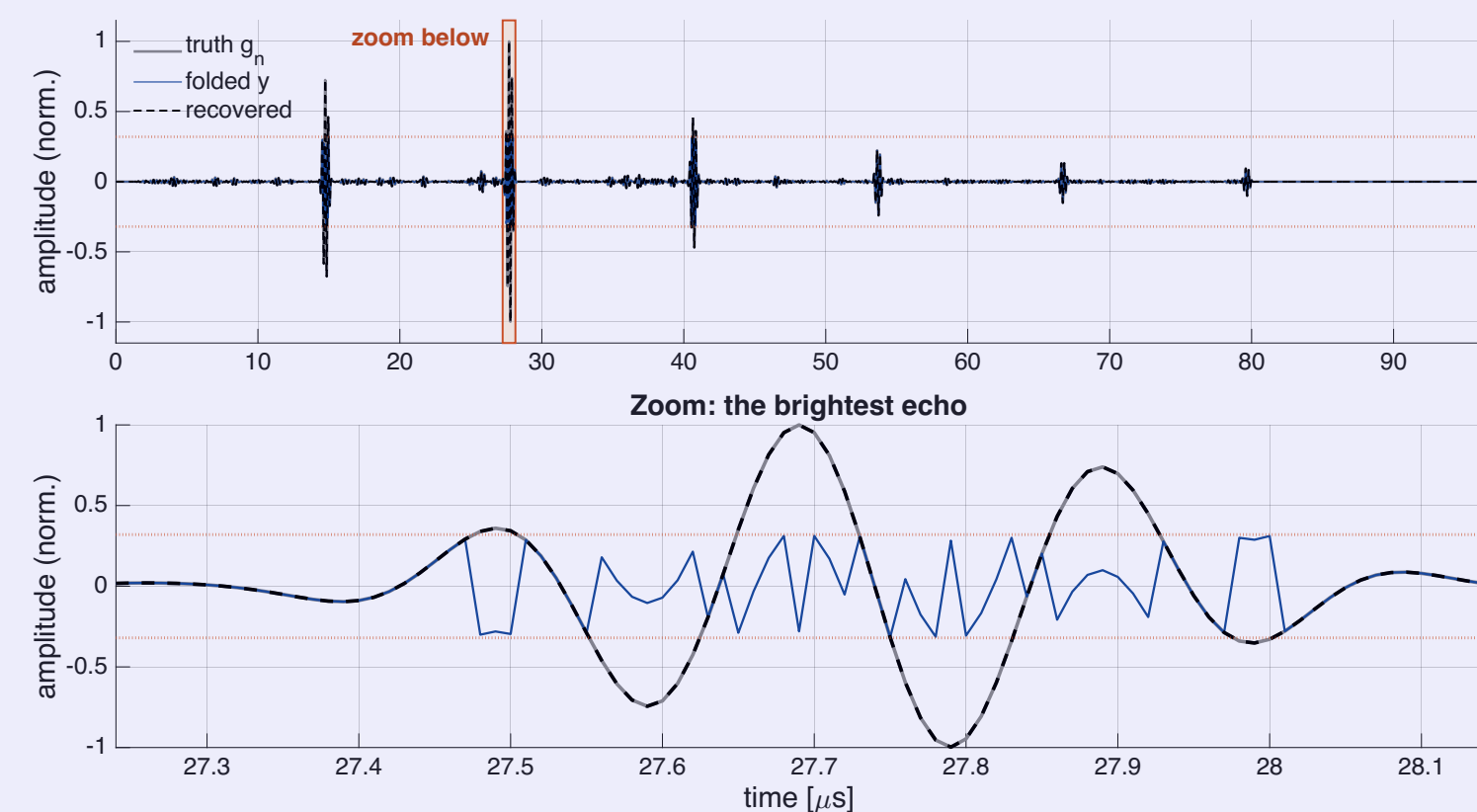


Preventing information loss from ADC saturation in ultrasound imaging

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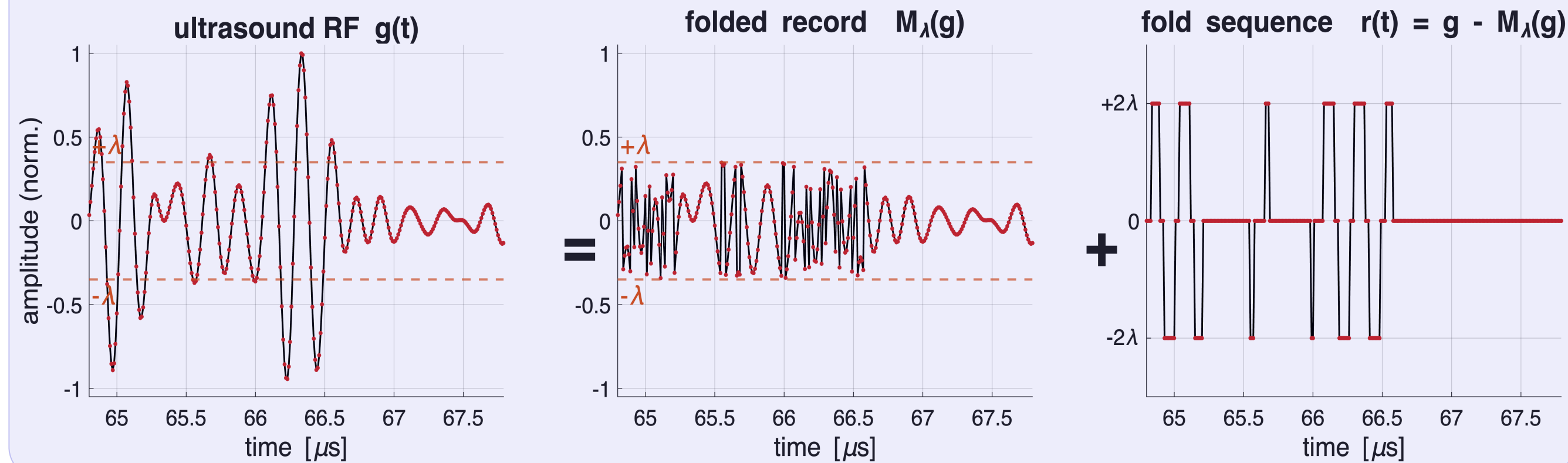
1 - The dynamic range (DR) problem of ultrasound (US)

- Echoes in a single US RF line can differ by 70 dB. [5]
- An ADC saturates on the bright echoes, causing clipping (information loss).



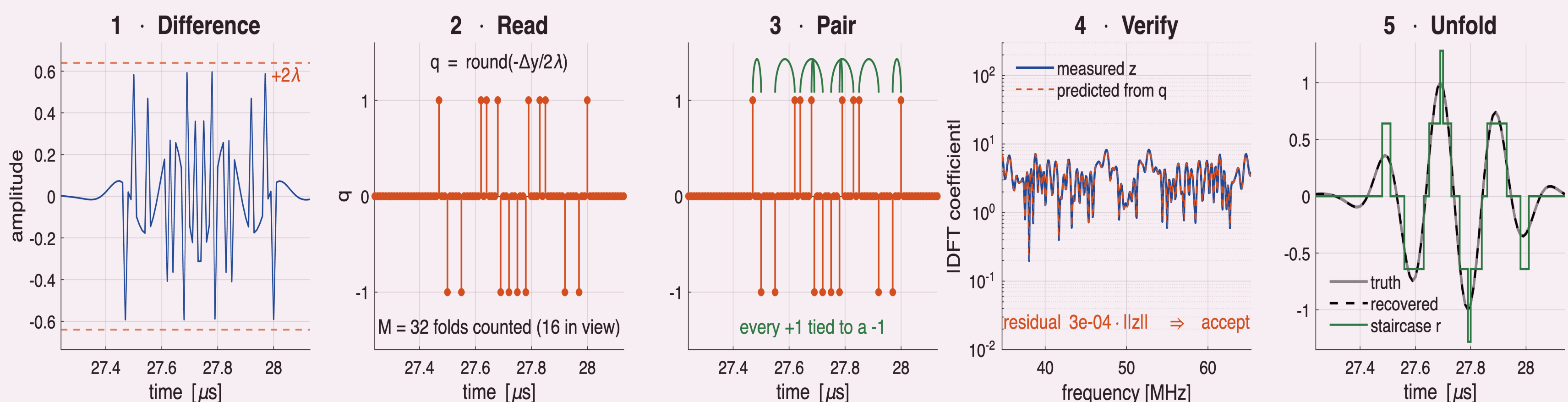
2 - The Unlimited Sensing Framework (USF) [1, 2] can circumvent information loss due to clipping

- A modulo ADC folds the waveform back into a lower (DR) $[-\lambda, \lambda]$. Recording folded samples y .
- The original signal g can be split as $g = y + r$.
- Recovering the sequence r algorithmically enables the recovery of g , avoiding the effects of clipping.



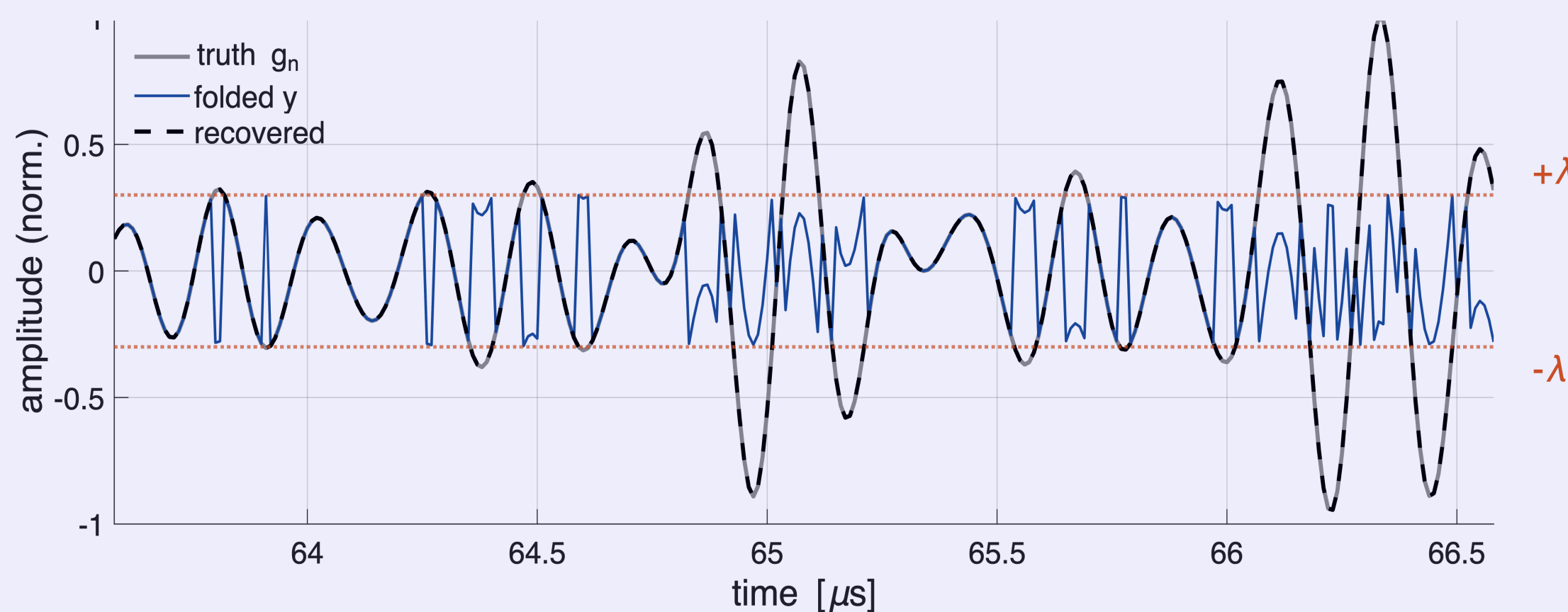
3 - Recovery algorithm

- The recovery uses the folded samples y and the threshold λ .
- First difference Δy : folds become sparse $\pm 2\lambda$ jumps
 - Read: count the jumps
 - Pair: each exit with its re-entry
 - Verify: out-of-band spectrum
 - Unfold: $\hat{g} = y + \hat{r}$
- [1]–[6]



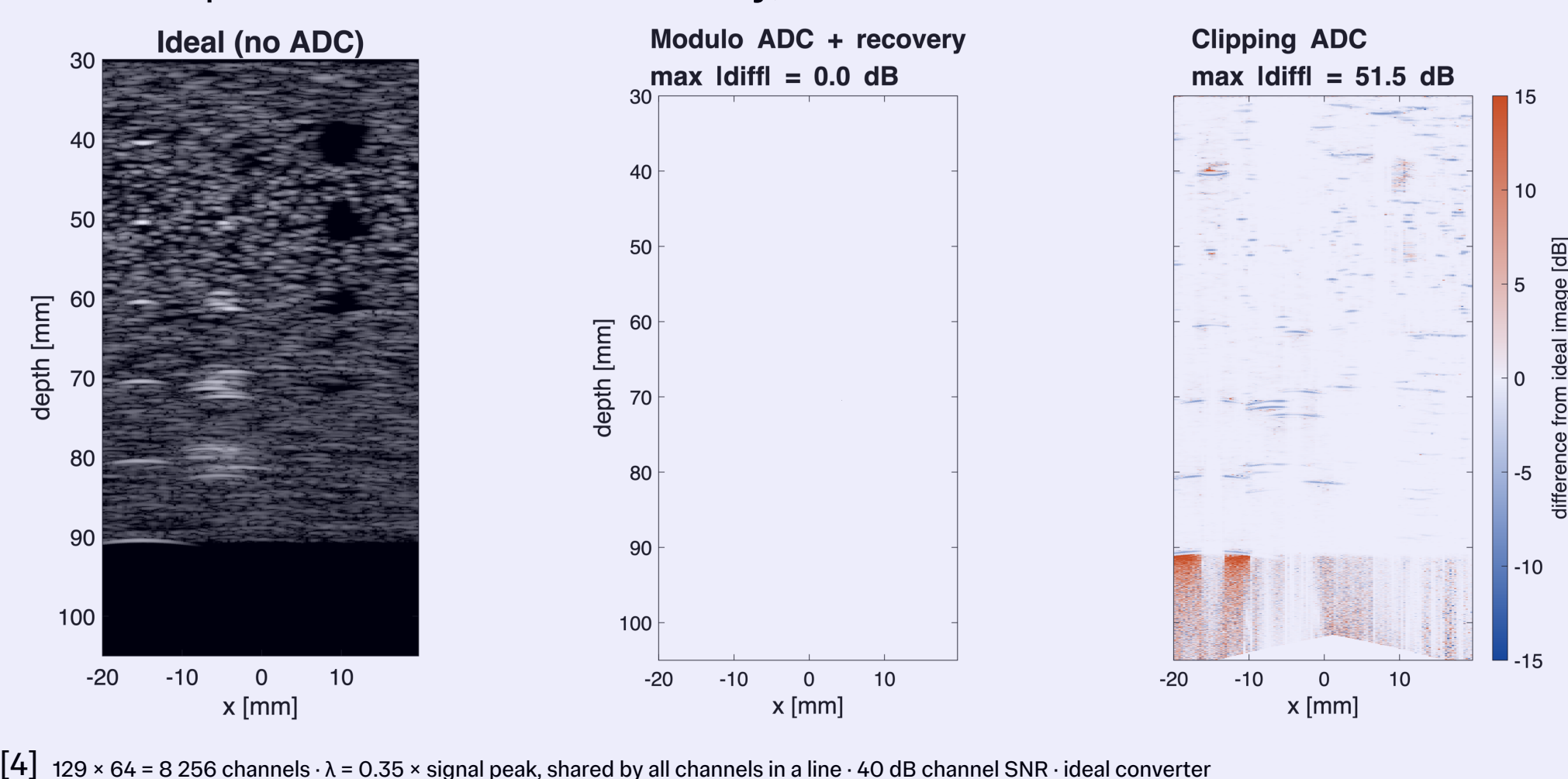
4 - Operating envelope

- Exact fold train in 50/50 noisy draws (80 dB SNR) down to $\lambda \approx 0.30$ of the peak. This yields recovered amplitudes of $\sim 3\times$ the rail λ (with a maximum-observed 256 folds).
- Exact as long as λ exceeds the steepest sample-to-sample difference of g : $\lambda > \max |\Delta g|$ [3].



5 - B-mode imaging after recovery

- The recovered amplitude is approx. $2.9\times\lambda$.
- Worst pixel error ≈ 0 dB after recovery, 51.5 dB if the same converter saturates.



6 - Implementation considering hardware artifacts

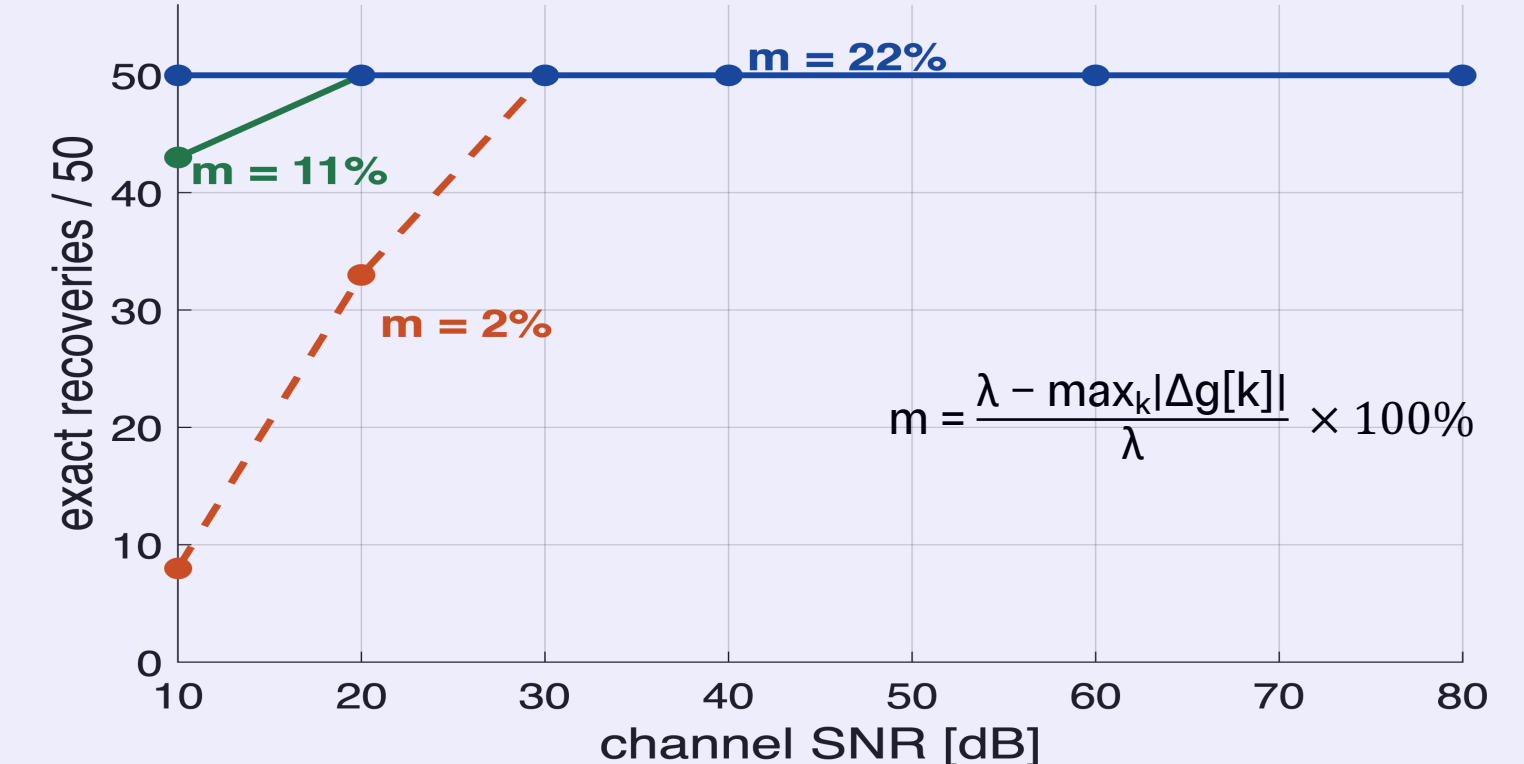
- The margin $m = \lambda - \max |\Delta g|$ is the “budget” that channel noise, quantisation noise of the ADC, and imperfect folds $2\lambda(q \pm \epsilon), \epsilon \sim U(-\epsilon_{\max}, +\epsilon_{\max})$ can spend, whilst allowing exact reconstruction of g .
- Noise: reconstruction exact from $\text{SNR} \geq 30$ dB.
- Bits: reconstruction error $\leq 1.4\%$ at 8–12 bits.
- Uncertain folding: recovered at up to 20% of 2λ , at moderate thresholds.
- All three together at λ value of 40% the signal's highest echo (20 dB, 8 bits, 5% jitter): exact in every test.

Takeaway

In simulation, this implementation of the USF recovers the full US signal amplitude from an ADC range spanning only a **third** of its tallest echo. There is a trade-off between folding to smaller λ , and the margin that guarantees recovery from channel noise, quantisation, and imperfect modulo hardware.

Limitations: US RF and recovery was simulation only, λ known, high sampling rates

Wider read margin $\rightarrow$ exact at lower SNR



Exact-recovery count vs channel SNR at three fold factors. Pre-fold Gaussian noise only (no quantisation nor fold jitter); $\lambda = \text{FF} \cdot \max |g|$ with $\text{FF} = 0.40, 0.35, 0.32$, giving read margins $m = 22\%, 11\%, 2\%$

References: [1] Bhandari, Krahmer, Raskar, IEEE TSP 2021. [2] Bhandari, Krahmer, Poskitt, IEEE TSP 2022. [3] Itoh, Appl. Opt. 1982. [4] Jensen, Field II, 1992/1996. [5] Szabo, Diagnostic Ultrasound Imaging, 2004. [6] Kvich, Yasar, Tasci, Yazicigil, Eldar, IEEE ICASSP 2025.